Algorithms, Data Structures, and AI Problem Solving

B03 - Exact 100-page production blueprint

Audience: Python learners ready to develop algorithmic judgment

Page architecture

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Chapter requirements

Chapter 1: Complexity and Measurement

Open with measurable objectives and prerequisites. Build intuition before formalism. Include at least two worked examples, an accessible visual or table, progressive practice, common mistakes, independent exercises, and retrieval review. Connect to executable Python or an interactive lab where appropriate.

Chapter 2: Arrays Strings and Two-Pointer Patterns

Open with measurable objectives and prerequisites. Build intuition before formalism. Include at least two worked examples, an accessible visual or table, progressive practice, common mistakes, independent exercises, and retrieval review. Connect to executable Python or an interactive lab where appropriate.

Chapter 3: Stacks Queues Hashing and Heaps

Open with measurable objectives and prerequisites. Build intuition before formalism. Include at least two worked examples, an accessible visual or table, progressive practice, common mistakes, independent exercises, and retrieval review. Connect to executable Python or an interactive lab where appropriate.

Chapter 4: Recursion Trees and Traversals

Open with measurable objectives and prerequisites. Build intuition before formalism. Include at least two worked examples, an accessible visual or table, progressive practice, common mistakes, independent exercises, and retrieval review. Connect to executable Python or an interactive lab where appropriate.

Chapter 5: Graphs and Network Reasoning

Open with measurable objectives and prerequisites. Build intuition before formalism. Include at least two worked examples, an accessible visual or table, progressive practice, common mistakes, independent exercises, and retrieval review. Connect to executable Python or an interactive lab where appropriate.

Chapter requirements - continued

Chapter 6: Sorting Searching and Selection

Open with measurable objectives and prerequisites. Build intuition before formalism. Include at least two worked examples, an accessible visual or table, progressive practice, common mistakes, independent exercises, and retrieval review. Connect to executable Python or an interactive lab where appropriate.

Chapter 7: Greedy Algorithms

Open with measurable objectives and prerequisites. Build intuition before formalism. Include at least two worked examples, an accessible visual or table, progressive practice, common mistakes, independent exercises, and retrieval review. Connect to executable Python or an interactive lab where appropriate.

Chapter 8: Dynamic Programming

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Chapter 9: Backtracking and Constraint Search

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Chapter 10: Algorithms Inside AI Systems

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Capstone

Build and benchmark a search-and-ranking pipeline using multiple data structures and algorithms.

Production and QA

Author original English content. Verify code, formulas, solutions, citations, diagrams, accessibility, and objective alignment. Render and reflow until the final book contains exactly 100 content pages without blank pages, repeated filler, oversized headings, or artificial spacing. Deliver source Markdown, plain text, PDF, answer guidance, and a QA report.